

2025 AMC 10A Problem 7 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 7. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 7

**Problem:**

Suppose a and b are real numbers. When the polynomial $x^3 + x^2 + ax + b$ is divided by $x - 1$, the remainder is 4. When the polynomial is divided by $x - 2$, the remainder is 6. What is $b - a$?

Answer Choices:

- A. 14
- B. 15
- C. 16
- D. 17
- E. 18

Solution:

When the polynomial $x^3 + x^2 + ax + b$ is divided by $x - 1$, the remainder is 4.

In other words,

$$x^3 + x^2 + ax + b = (x - 1)q(x) + 4$$

When $x = 1$, we obtain

$$a + b = 2$$

Similarly, When the polynomial is divided by $x - 2$, the remainder is 6.

In other words,

$$x^3 + x^2 + ax + b = (x - 2)r(x) + 6.$$

When $x = 2$, we obtain

$$2a + b = -6$$

However, we know from above that $a + b = 2$, which gives us $a = -8$ and $b = 10$.

Thus,

$$b - a = 10 - (-8) = \boxed{\text{(E)} 18}.$$

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